**Experiment 9 **

Q1 Answer following question in typed form and paste it in the text box provided for submission:

1 When do we go for Chi square test ?

We go for a Chi-Square (χ^2) test when we want to analyze categorical data and check whether there is a relationship or difference between categories.

When to use Chi-Square Test To test relationship between two categorical variables Used in Chi-Square Test of Independence. Example: Check whether gender is related to product preference. **To test whether observed data fits expected data** Used in Chi-Square Goodness-of-Fit Test. Example: Check whether dice outcomes follow expected probabilities. **When data is in frequency form (counts)** Data should be number of occurrences, not percentages or averages. Example: Number of students choosing Science, Commerce, or Arts. **When variables are categorical** Such as male/female, yes/no, pass/fail, category types. **When sample size is reasonably large** Expected frequency in each category should usually be ≥ 5.

Smple Examplei

Suppose we want to test whether study method affects exam result.

Study Method	Pass	Fail
Online	30	10
Offline	20	20

We use Chi-Square Test to check if study method and result are independent or related.

2 Do we use Chi square test for independence ? Explain How ?

Chi-Square test is used for testing Independence of two categorical variables.

The Chi-Square Test of Independence checks whether two categorical variables are related or independent of each other.

Example:

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Gender and Product Preference Study Method and Exam Result Education Level and Job Type

If variables are independent, one variable does not affect the other.

Steps to Perform Chi-Square Test of Independence State Hypothesis H_0 (Null Hypothesis): Variables are independent (no relationship). H_1 (Alternative Hypothesis): Variables are dependent (there is a relationship). Create Contingency Table Arrange observed frequencies in a table.

Example:

Study Method	Pass	Fail	Total
Online	30	10	40
Offline	20	20	40
Total	50	30	80

Calculate Expected Frequency

Formula:

E

$(Row$

$Total \times Column$

$Total)Grand$

$Total$ E= Grand Total (Row Total×Column Total)

Example: Expected value for Online−Pass

E

40 × 50 80

25 E= 80 40×50

=25 Calculate Chi-Square Value χ 2

$\sum(O-E)2E$ χ 2 =∑ E (O−E) 2

Where

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O = Observed frequency E = Expected frequency

Find Degree of Freedom df

$(r-1)(c-1)$ df=(r−1)(c−1)

where

r = number of rows c = number of columns Decision Rule Compare calculated χ^2 value with critical χ^2 value from Chi-square table. Or check p-value. If χ^2 calculated > χ^2 table value → Reject H_0 If χ^2 calculated ≤ χ^2 table value → Accept H_0

3 Explain role of Critical value and p-value in Chi square test.

Role of Critical Value and p-value in Chi-Square Test

In a Chi-Square (χ^2) test, the critical value and p-value help us decide whether to accept or reject the null hypothesis (H_0).

1. Critical Value

Definition: The critical value is the value taken from the Chi-Square distribution table based on:

Significance level (α) (commonly 0.05) Degrees of freedom (df)

Role in Chi-Square Test:

It acts as a cut-off point for decision making. We compare the calculated χ^2 value with the critical value.

Decision Rule:

If χ^2 calculated > χ^2 critical value → Reject H_0 If χ^2 calculated ≤ χ^2 critical value → Accept H_0

Meaning: If the calculated value is larger than the critical value, it means the difference between observed and expected frequencies is significant.

2. p-value

Definition: The p-value is the probability of obtaining the observed result assuming the null hypothesis is true.

Role in Chi-Square Test:

It shows the strength of evidence against H_0. It is compared with the significance level (α).

Decision Rule:

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If p-value ≤ α (0.05) → Reject H_0 If p-value > α → Accept H_0

Meaning: A small p-value indicates that the observed difference is unlikely to occur by chance, so the variables may be related.

Simple Example

Suppose:

χ^2 calculated = 10.5 Critical value = 7.81 p-value = 0.02 α = 0.05

Decision:

10.5 > 7.81 → Reject H_0 0.02 < 0.05 → Reject H_0

4 Can we use chi square test to show existence of correlation between 2 features in a dataset ? How will you interpret test result to indicate correlation presence. **bold text**

Chi-Square test can be used to check whether there is a relationship (association) between two categorical features in a dataset. If a relationship exists, it indicates correlation/association between the features.

When we use it When both features are categorical (e.g., Gender, Education level, Product type). When data is in the form of frequency counts.

Steps to Check Correlation Using Chi-Square Test **Form Hypotheses** H_0 (Null Hypothesis): The two features are independent (no correlation). H_1 (Alternative Hypothesis): The two features are dependent (correlated/associated). **Create a Contingency Table** Example:

Gender	Like Product	Do Not Like
Male	30	20
Female	40	10

Calculate Expected Frequencies E

$(Row$

$Total \times Column$

$Total)Grand$

$Total$ E= Grand Total (Row Total×Column Total)

Calculate Chi-Square Statistic $\chi 2$

$\sum(O-E)2E$ $\chi 2 = \sum$ E (O−E) 2

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Where

O = Observed frequency E = Expected frequency Find p-value or compare with Critical Value

Interpretation of Test Result

Case 1: p-value ≤ 0.05 (or χ^2 calculated > χ^2 critical)

Reject H_0 The two features are not independent Correlation / association exists between the features

Case 2: p-value > 0.05 (or χ^2 calculated ≤ χ^2 critical)

Accept H_0 The two features are independent No correlation between the features Important Note Chi-square shows association, not the strength or direction of correlation like Pearson correlation

5 What is degrees of freedom and how to calculate it while performing chi square test. **bold text**

Degrees of Freedom (df)

Degrees of Freedom refers to the number of values that are free to vary in the calculation after certain constraints are applied.

In a Chi-Square test, degrees of freedom help determine the correct critical value from the Chi-Square distribution table for hypothesis testing.

How to Calculate Degrees of Freedom in Chi-Square Test

For a Chi-Square Test of Independence, the formula is:

$$df$$

$(r-1)(c-1)$ df=(r−1)(c−1)

Where:

r = number of rows c = number of columns

Example

Suppose we have the following contingency table:

Study Method	Pass	Fail
Online	30	10

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Study Method	Pass	Fail
Offline	20	20
Hybrid	25	15

Number of rows (r) = 3 Number of columns (c) = 2 d f

(3 − 1) (2 − 1) df=(3−1)(2−1) d f

2 × 1

2 df=2×1=2

So, degrees of freedom = 2.

Role of Degrees of Freedom It determines the shape of the Chi-Square distribution. It is used to find the critical value from the Chi-Square table. It helps in deciding whether to accept or reject the null hypothesis.

Close

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**Conclusion**
The Chi-Square test is used to determine
association between categorical variables
with the expected frequencies.
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Conclusion The Chi-Square test is used to determine whether there is a significant association between categorical variables. It compares the observed frequencies with the expected frequencies.

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